

FRANSTA, Sweden

WMO: 023420

Lat: 62.517N

Lon: 16.183E

Elev: 344

StdP: 14.51

Time Zone: 1.00 (EUC)

Period: 82-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.6	-15.3	-24.9	1.4	-21.4	-18.7	2.0	-15.1	22.9	34.8	19.6	27.9	0.1	20	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	79.4	62.6	75.4	61.0	71.3	58.7	64.9	75.7	62.5	72.0	60.2	68.6	6.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.6	79.9	68.4	58.3	73.5	66.1	56.4	68.6	64.0	30.1	75.8	28.3	72.3	26.7	68.7	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.1	15.8	12.1	-26.9	85.1	8.1	3.3	-32.7	87.4	-37.4	89.3	-42.0	91.1	-47.9	93.5		
			-27.1	67.8	8.0	2.9	-32.8	69.8	-37.5	71.5	-42.0	73.2	-47.8	75.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	13.2	17.7	27.8	35.8	47.0	54.8	59.2	55.5	46.6	37.5	26.2	17.4
	DBStd	18.00	15.59	13.59	8.22	6.47	6.65	6.23	5.17	5.22	5.40	7.40	9.83	12.92
	HDD50	5578	1141	906	688	426	142	22	1	9	131	387	713	1012
	HDD65	10365	1606	1326	1153	875	558	313	193	297	553	851	1163	1477
	CDD50	710	0	0	0	1	49	164	286	181	28	1	0	0
	CDD65	23	0	0	0	0	0	6	13	4	0	0	0	0
	CDH74	490	0	0	0	0	39	129	260	61	1	0	0	0
	CDH80	90	0	0	0	0	2	30	47	11	0	0	0	0
	WSAvg	3.7	2.7	2.7	4.1	4.2	4.6	4.7	4.6	3.7	3.7	3.7	2.8	2.7
	PrecAvg	23.5	1.5	0.9	1.0	1.3	1.7	2.6	3.6	3.3	2.0	2.2	1.8	1.6
(15) Precipitation	PrecMax	34.5	2.8	2.2	2.2	3.6	3.9	4.8	9.4	5.9	5.5	5.9	4.0	2.9
	PrecMin	17.5	0.2	0.1	0.2	0.1	0.5	0.8	1.0	1.4	0.6	0.6	0.5	0.2
	PrecStd	4.1	0.7	0.6	0.5	0.9	0.8	1.0	2.1	1.5	1.1	1.3	0.9	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.9	45.2	49.4	66.5	78.6	84.4	84.5	81.3	68.2	57.8	49.2	41.3
		MCWB	38.7	39.3	41.9	52.1	57.5	66.8	65.8	64.8	56.9	51.8	44.3	37.8
	2%	DB	39.2	41.1	44.7	57.5	73.3	77.3	80.2	74.2	63.1	54.0	44.1	38.6
		MCWB	35.2	36.3	38.0	46.4	56.4	62.2	62.9	62.0	54.5	48.5	41.2	35.9
	5%	DB	35.4	37.9	41.7	51.8	67.7	72.4	76.5	69.6	59.3	51.2	40.9	36.1
		MCWB	32.6	34.8	36.4	42.5	54.3	58.8	61.6	58.8	52.6	47.5	38.6	34.0
	10%	DB	32.7	34.9	38.8	47.9	62.7	68.4	72.0	66.3	56.1	48.7	38.2	33.5
		MCWB	30.9	32.5	34.3	40.5	50.9	56.7	59.7	57.4	50.3	45.7	36.3	32.2
	0.4%	WB	39.4	40.2	42.2	52.4	60.9	67.1	67.4	66.4	58.9	53.0	44.8	38.4
		MCDB	43.2	44.4	48.2	66.3	73.9	80.9	80.0	77.4	64.8	55.6	47.6	40.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.7	37.2	39.3	47.2	57.6	64.0	65.3	63.2	56.2	50.4	41.4	36.4
		MCDB	38.5	40.2	43.3	56.0	70.2	75.6	76.3	71.3	61.4	52.5	43.9	38.1
	5%	WB	33.0	34.9	36.8	43.6	54.8	60.3	62.9	60.9	53.5	48.0	38.8	34.3
		MCDB	34.9	37.4	40.6	50.2	66.8	69.9	73.0	67.6	57.9	50.4	40.4	35.8
	10%	WB	31.0	32.7	35.0	41.2	51.7	57.9	60.7	58.6	51.5	45.8	36.5	32.2
		MCDB	32.7	34.7	38.0	46.2	60.8	66.7	69.9	63.9	54.8	48.7	38.1	33.4
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	17.8	16.8	18.1	22.5	19.6	20.0	17.6	16.5	13.7	11.3	13.6
		MCDBR	15.1	15.8	19.6	27.5	32.3	28.1	29.6	23.9	21.8	17.6	11.8	13.4
		MCWBR	13.1	12.5	13.5	16.6	16.3	14.2	13.8	11.6	12.9	12.3	9.8	11.5
	5% WB	MCDBR	14.4	14.4	16.9	24.6	30.1	26.0	25.2	21.0	17.6	14.0	11.2	13.1
		MCWBR	13.0	12.0	12.2	15.9	15.5	14.5	13.6	11.5	11.3	11.0	9.9	11.6
(40) Clear-Sky Solar Irradiance	taub		0.230	0.287	0.313	0.350	0.354	0.349	0.365	0.356	0.323	0.308	0.247	0.347
	taud		2.090	2.233	2.311	2.371	2.426	2.470	2.443	2.461	2.542	2.455	2.177	2.652
	Ebn at Noon		139	211	252	265	274	278	269	261	251	207	132	87
	Edh at Noon		10	20	26	31	32	31	31	28	21	16	9	6
(44) All-Sky Solar Radiation	RadAvg		64	266	730	1203	1505	1735	1581	1203	755	331	94	34
	RadStd		8	41	78	101	154	117	178	102	91	57	18	3

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page